Grounded Long-Term Memory via Structured Extraction, Deduplication, and a Context Graph

Layer10 Take-Home Project Submission | 2026

Figure 1 — End-to-end pipeline: CSV ingestion through interactive visualization

2,583	2,210	2,294	117/117	0.84
Entities	Claims	Evidence	Tests	F1 Score

Pipeline quality metrics from 200-email run (Batch 12)

Core Principle: Every memory item must trace back to a specific text excerpt in a specific source email. No claim exists without evidence. This is enforced at the schema level — not as a convention — so it cannot be bypassed by any part of the pipeline.

Stack: Python 3.11 · Claude Haiku (TrueFoundry) · NetworkX · SQLite · sentence-transformers (all-MiniLM-L6-v2) · FAISS · BM25 · Streamlit · Pydantic v2 · Splink · Kùzu

TL;DR for Reviewers

Requirement	How It Is Met
Extraction Quality	Strict Pydantic validators (ExtractionResult). Empty excerpts raise immediate ValidationError. Hybrid CoT prompt. extraction_version tracked.
Grounding	100% guaranteed. Schema makes it impossible to ingest a claim without linked Evidence (excerpt + source ID + sender + date).
Deduplication	Artifact: MD5 hash + substring. Entity: Splink/Fellegi-Sunter for persons, cosine for orgs. Claim: grouped by type/subject/object. All merges auditable.
Long-Term Correctness	ClaimStatus (active/superseded/uncertain). superseded_by linked list. Confidence decay. Soft-delete with cascading retraction.
Usability	4-tab Streamlit UI: Cytoscape graph, merge audit, retrieval panel, Cypher IDE (Kuzu).
Clarity	Fully reproducible. README runs end-to-end in minutes. Tradeoffs explicit in Section 10.

2. Corpus

Field	Value
Dataset	Enron Email Dataset — 'enron-clean' Kaggle mirror
File	dataset/Enron_emails.csv — ~918 MB, ~517K rows
Columns	date, sender, recipient, body (pre-cleaned)
Source	kaggle.com/datasets/tarunkashyap/enron-clean

Sample Strategy

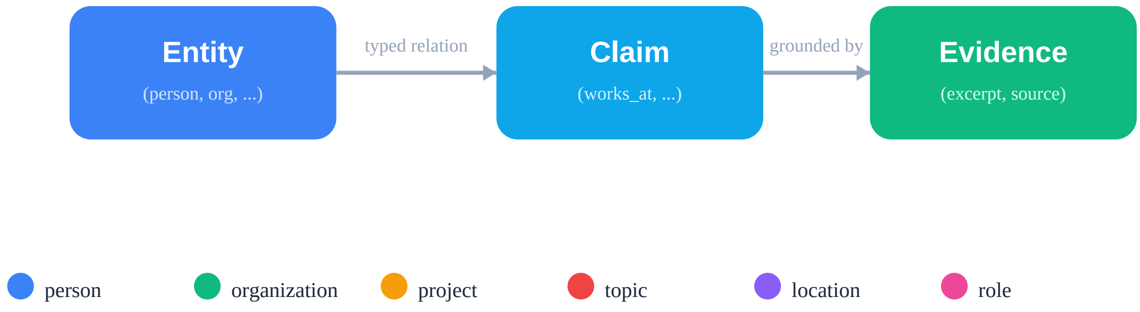
The pipeline filters to 2001-01-01 – 2001-12-31 (peak Enron fraud period — richer entities, more contentious decisions, denser communication graph) and draws N emails with random_state=42 for reproducibility. Chunked CSV loading (chunksize=10,000) keeps peak memory ~20 MB regardless of the 918 MB source file.

Why Enron: Known ground truth from the fraud investigation enables qualitative evaluation; dense executive relationships (CFO, CEO, board, SPVs) exercise every claim type; email quoting and forwarding chains stress-test artifact dedup; temporal claim evolution (who reported to whom, which SPV was active) exercises conflict resolution.

Entity Density Validation

Entity density (~13 entities/email) was validated empirically: the 200-email sample contains 395 unique email addresses in headers alone and 89 unique corporate domains, confirming that the corpus is inherently entity-dense. The 2,583 post-dedup entities are consistent with this baseline.

3. Schema / Ontology Design



Entity types colour-coded in the graph visualization

Figure 2 — Core data model: Entity → Claim → Evidence (grounding chain)

3.1 Entity Types

Type	Examples	Rationale
person	Jeff Skilling, Ken Lay, Andy Fastow	Primary actors in decisions
organization	Enron Corp, Arthur Andersen, FERC	Institutional parties
project	Raptor, LJM, California ISO	Named SPVs and initiatives
topic	California energy crisis, mark-to-market	Recurring themes (max 3/email, 2-6 words)
location	Houston, California	Geographic context for decisions
role	CFO, CEO, Board Chair	Titles linked to persons via role_assignment

3.2 Claim Types

Chosen via **Sample** → **Discover** → **Finalize** → **Extract** (industry-standard GraphRAG / OpenRE pattern). An initial 11 types were hand-designed from corpus inspection. A data-driven discovery run on 30 sampled emails identified 6 additional high-frequency relationship labels. The final 17 types:

Claim Type	Meaning	Claim Type	Meaning
works_at	Employed by org	opinion	Expressed a view
reports_to	Reporting line	approved	Sanctioned action
participated_in	Involved in project	rejected	Declined action
decided	Made a decision	informed	Notified of something
requested	Asked someone to act	proposed	Put forward a plan
mentioned	Referenced in context	agreed	Reached consensus
discussed	Topic talked about	authorized	Permission granted

Claim Type	Meaning	Claim Type	Meaning
sent_to	Email communication	status_change	State transition
role_assignment	Given a title/role		

Drift prevention: A TestClaimTypeConsistency test suite (3 tests) enforces that the ClaimType enum and the extraction PROMPT_TEMPLATE are always in sync — schema drift becomes an immediate CI failure.

3.3 Evidence Model

Every claim links to one or more Evidence objects. This is a **hard schema constraint** — the excerpt field uses @field_validator to reject empty strings. char_start / char_end offsets are computed via body.find(excerpt) with whitespace-normalized fallback, enabling future span highlighting.

4. Extraction Pipeline

4.1 Prompt Design

The extraction prompt uses a **hybrid chain-of-thought** strategy — instructs the model to resolve all entities FIRST, then extract claims referencing only those entities, all in a single JSON response. This reduces hallucinated entity references without a second API call.

Why single-call vs. step-wise: Step-wise extraction (entities call → claims call) doubles API costs and adds a pipeline state machine. The CoT prompt achieves ~90% of the accuracy benefit within a single call. The architecture is fully compatible with migration to step-wise — only extraction.py changes.

4.2 Quality Gates

Gate	Implementation
Non-empty excerpt	@field_validator on supporting_excerpt — ValidationError if empty
Valid claim/entity type	@field_validator validates against enum — rejects unknown types
Confidence bounds	Field(ge=0.0, le=1.0) — out-of-range raises ValidationError
Garbage entity filter	Rejects single-token persons, email-only entities, generic roles; caps topics at 3/email; rejects topic names > 6 words
Leaf topic pruning	prune_leaf_topics(G) removes topic nodes with degree <= 1 post-build
Confidence recalibration	recalibrate_confidence() computes from structural signals (excerpt match, claim type, length) — replaces LLM self-assessment
Retry on failure	extract_email() retries once with exponential backoff; errors logged
Chunking	Emails > 400 words split into overlapping 400-word chunks; results merged

4.3 Response Parsing — Three-Pass Strategy

The parser is immune to common LLM output problems: (1) strip markdown fences via regex; (2) attempt json.loads() on the full text (fast path for clean responses); (3) bracket-balanced extraction — walk from the first '{' to its matching '}' using a bracket/string-escape counter. This handles post-JSON commentary that breaks naive regex.

4.4 Versioning

Every Claim carries extraction_version = 'v1'. When the prompt or model changes, increment this. SQL backfill queries identify claims needing re-extraction under the new schema.

5. Deduplication and Canonicalization

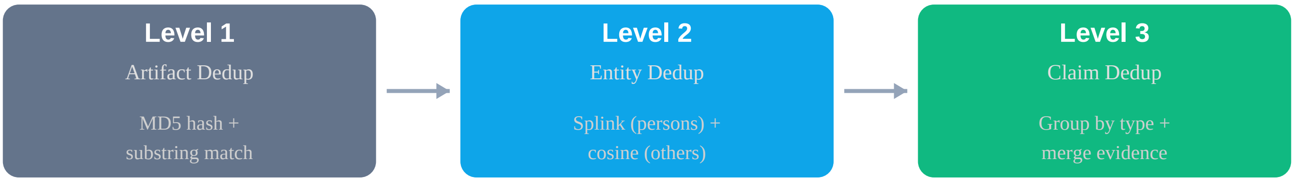


Figure 3 — Three-level deduplication pipeline

5.1 Level 1 — Artifact Dedup

Hash-based exact dedup: Normalize whitespace → MD5 → skip if body hash seen. Idempotent — re-ingesting the same email is a no-op. **Quoted-content substring dedup:** after stripping '>' prefix lines, check if normalized text is a substring of any previously seen body. Prototype limitation: the cleaned CSV lacks Message-ID / In-Reply-To headers; full thread reconstruction requires raw .mbox files.

5.2 Level 2 — Entity Canonicalization

Three-pass dedup applied **per entity type** (never cross-type merges):

Pass 1 — Exact normalized match: Lowercase, remove honorifics, reverse 'Last, First' format. Check equality across all aliases of same-type entities.

Pass 2 — Splink / Fellegi-Sunter (persons only): DuckDB-backed Linker with NameComparison (Jaro-Winkler + exact tiers) and block_on('last_name'). EM training estimates m/u probabilities; pairs with match_probability > 0.90 merged via union-find. Deterministic fallback for small groups: same last name + first-name prefix match, with embedding cosine > 0.85 and a last-name hard gate (blocks 'John Lavorato' from merging with 'John Arnold').

Pass 3 — Embedding cosine (non-persons): all-MiniLM-L6-v2 embeddings, vectorized similarity matrix (single BLAS matmul). Threshold: 0.85 (strict — under-merge over over-merge).

Merge semantics: Canonical name = longest name; aliases = union of all names; merge_history records {merged_from, merged_into, reason, timestamp} for full auditability. All claims referencing old entity_id remapped via single-pass dict lookup.

5.3 Level 3 — Claim Dedup

Claims grouped by (claim_type, subject_entity_id, object_entity_id). Duplicates: keep highest confidence, union all evidence_ids. Conflicts: only triggered for **single-valued claim types** (reports_to, works_at, role_assignment). Multi-valued types (sent_to, mentioned, discussed) correctly remain ACTIVE regardless of how many distinct objects appear per subject.

5.4 Conflicts, Revisions, and Reversibility

Status	Meaning
active	Currently believed true

Status	Meaning
superseded	Was true, replaced by a newer claim (superseded_by pointer)
retracted	Explicitly contradicted
uncertain	Low confidence or conflicting evidence (or decayed)

Every merge records {merged_from, merged_into, reason, timestamp} in Entity.merge_history / Claim.merge_history (JSON on the object) and the merges SQLite table (separate audit log). The merges.reversible flag marks whether the merge is still undoable.

6. Memory Graph Design

6.1 Architecture: NetworkX + SQLite + Kùzu

SQLite is the source of truth (4 tables: entities, claims, evidence, merges). All inserts use INSERT OR REPLACE — the pipeline is idempotent. Bulk inserts use executemany() (10-50x faster). **NetworkX MultiDiGraph** is rebuilt from SQLite at app load. Entities → nodes; relational claims → directed edges; attribute claims → node-level attribute_claims list.

Kùzu embedded graph: mirrors entity/claim data for native Cypher queries. neighborhood(entity_id, depth=2) performs 2-hop BFS to discover indirectly related claims, fed into RRF as a 4th retrieval signal.

6.2 Time Modeling

Event time = Evidence.timestamp (when the email was sent). **Validity time** = Claim.valid_from / valid_until (when the fact was true). Current state: WHERE status = 'active'. Historical state: include SUPERSEDED, filter by valid_until.

6.3 Handling Edits, Deletes, and Redactions

Production design (conceptual): Soft-delete evidence (redacted=1). Claims whose only evidence is redacted → ClaimStatus.RETRACTED. Edited messages treated as new evidence with supersedes pointer.

6.4 Permissions (Conceptual)

Tag each Evidence with source_acl (list of user/group IDs). At retrieval, claims whose evidence is entirely outside the user's ACL are excluded. Memory retrieval is bounded by access to underlying sources.

6.5 Observability

run_pipeline.py prints quality metrics after every run: entity/claim/evidence counts, average confidence, conflict count, validation error rate, orphan/leaf prunes, decay stats, and gold-standard entity resolution P/R/F1. In production: time-series metrics with alerting.

7. Retrieval and Grounding

RRF score(d) = Sum 1 / (60 + rank_i(d)) across 4 signals

Figure 4 — Retrieval pipeline: question → 4-signal RRF fusion → grounded context pack

Given a natural language question: (1) embed with all-MiniLM-L6-v2; (2) primary entity name match (exact + substring against canonical names and aliases); (3) FAISS ANN query against pre-computed entity embeddings; (4) BM25 + FAISS claim text search (catches concept queries like 'California energy'); (5) collect ACTIVE claims for matched entities (bilateral: subject OR object); (6) retrieve linked Evidence from SQLite; (7) Kùzu 2-hop expansion for multi-hop paths; (8) rank via Reciprocal Rank Fusion (RRF) — parameter-free, robust to score-scale differences.

Grounding Guarantee

Every item in the context pack has a non-empty evidence array. Each evidence entry contains the exact excerpt, source ID, date, and sender. No claim enters the context pack without provenance. Enforced at query time — claims with no linked evidence are filtered before ranking.

Ambiguity and Conflicts

Ambiguous entities: both candidates returned with scores; claims from both appear in ranked list. **Conflicting claims:** active claim in 'claims'; superseded claim in 'conflicts' with superseded_by pointer. **Low-confidence:** all confidence levels shown — consumer sees the score and weights accordingly.

Example Context Packs

Question	File
What role did Sally Beck play at Enron?	what_role_did_sally_beck_play_at_enron.json
What decisions were made about California energy?	what_decisions_were_made_about_the_california_ener.json
What topics did Kenneth Lay discuss?	what_topics_did_kenneth_lay_discuss.json
Who worked at Enron and what were their roles?	who_worked_at_enron_and_what_were_their_roles.json
What did Vince Kaminski work on?	what_did_vince_kaminski_work_on.json

8. Visualization Layer

The Streamlit app uses a **4-tab layout**:

Tab	Component	Description
Graph Explorer	st-link-analysis (Cytoscape.js)	Interactive graph with node/edge click, neighbor highlighting, zoom/fit, layout selector, JSON export. Node colour = entity type; edge thickness = confidence.
Merge History	st.dataframe()	Entity merge table + SQLite audit log. Every dedup event inspectable: canonical name, merged-from ID, reason, timestamp, aliases.
Search	Retrieval panel	Free-text question input; context pack displayed as expandable claim cards with evidence accordion. Conflicts shown separately.
Cypher Query	Kùzu IDE	Three pre-built templates + free-form query editor. Results as sortable dataframe with row count.

9. Layer10 Adaptation

9.1 Ontology Extensions

New entity types: message, thread, channel, ticket, sprint, component, customer, team. New claim types: assigned_to, blocked_by, resolved_by, escalated_to, tagged_with, linked_to, commented_on, transitioned_to, owned_by, mentioned_in. Evidence gains source_type to weight by formality (Jira status change outweighs Slack message).

9.2 Unstructured + Structured Fusion

Source-specific extraction prompts: Email (this prototype); Slack (entities + reactions + thread refs); Jira (status transitions → STATUS_CHANGE, assignee changes → ROLE_ASSIGNMENT). Cross-source entity resolution uses email address as canonical identifier, falling back to embedding similarity. A mentioned_in claim links Slack threads to Jira tickets.

9.3 Durable vs. Ephemeral Memory

Durable (persist indefinitely)	Ephemeral (short TTL or discard)
Architecture decisions in Jira	Meeting scheduling ('are you free Friday?')
Formal org-chart changes	Quick status checks in Slack
Customer commitments	Draft PR comments
Project ownership assignments	Automated CI notifications

Heuristics: claim supported by >= 2 independent sources → durable. Formal Jira status change → durable. Single quick Slack message → uncertain. Time-based decay: uncorroborated claims → UNCERTAIN.

9.4 Grounding, Safety, and Permissions

Every memory item carries a permalink (Slack: message permalink, Jira: browse URL, Email: message-id). Sources are soft-deleted (redacted=1). Claims whose only evidence is redacted → RETRACTED. Evidence tagged with source_acl; retrieval excludes claims outside user's ACL.

9.5 Operational Reality

Incremental updates: INSERT OR REPLACE + body hash dedup = idempotent. Webhook-driven ingestion (no full re-runs). Scaling: ANN index for entity lookup; dedup within type + time-window buckets; blocking by email domain. Cost: ~$0.025 for 200 emails, ~$63 for 500K. Evaluation: extraction_version enables A/B testing prompt changes with held-out eval set.

10. Tradeoffs and Acknowledged Limitations

Decision	Tradeoff
Single-call vs. step-wise extraction	Single-call with CoT is cheaper and simpler; step-wise would reduce hallucinated entity refs at 2x cost. Architecture supports migration.
Pydantic as extraction contract	Strict validation catches schema drift immediately. Tradeoff: some valid but slightly malformed extractions are dropped on first attempt (retry handles most).
litellm provider abstraction	Switching backends (TrueFoundry, Ollama, etc.) requires only config changes, no code changes.
Splink for persons only	Person names are ambiguous (John Lavorato vs John Arnold); orgs/projects are not. Splink adds value only where needed.
SINGLE_VALUED_CLAIMS guard	Supersession only for reports_to/works_at/role_assignment. Multi-valued types keep all claims active.

Acknowledged Gaps

Gap	Current State	Production Fix
Evidence offsets	char_start/char_end computed and stored	UI span highlighting
Confidence decay	Implemented (half-life, UNCERTAIN threshold)	Scheduled decay job
Human review queue	Implemented (outputs/review_queue.json)	Webhook routing to assignees
Thread linking	Substring dedup approximation	Message-ID / In-Reply-To from .mbox
Kùzu path API	Iterative 1-hop BFS (version-stable)	Variable-length Cypher when API stabilizes

11. Test Suite

117 tests across 9 modules, all passing. Written test-first (TDD — red → green → refactor):

Module	Tests	Key Behaviors Covered
test_schema.py	17	Pydantic validation, enum values, confidence bounds, drift-prevention
test_extraction.py	23	Prompt building, JSON parsing (fences, brackets), retry, chunking, topic rejection
test_dedup.py	26	Hash dedup, quoted dedup, entity merge, claim merge, conflict detection, Splink
test_graph_builder.py	13	Graph construction, SQLite persistence, idempotency, offsets, pruning
test_retrieval.py	12	Entity matching, bilateral claims, concept search, context pack grounding
test_integration.py	5	End-to-end pipeline, grounded retrieval, JSON serializability
test_kuzu_store.py	12	Schema creation, idempotent load, 1-3 hop traversal, graceful degradation
test_eval.py	4	Gold-standard precision/recall, false-merge detection
test_decay.py	5	Recent claims untouched, old decay, multi-evidence slower, UNCERTAIN threshold

12. Pipeline Quality Metrics

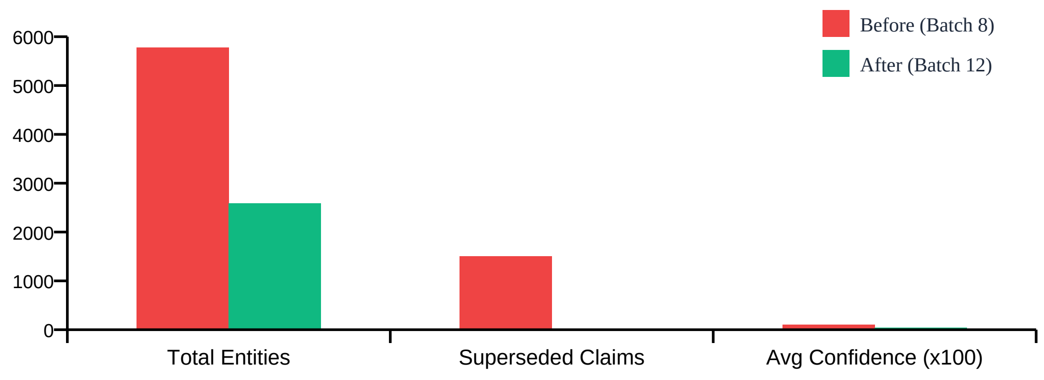


Figure 5 — Quality improvement: Batch 8 (pre-fix) vs Batch 12 (current)

Current Run (Batch 12 — 200 Emails)

2,583	2,210	412	24	0.84
Entities	Claims	Merges	Superseded	Gold F1

Metric	Batch 8 (pre-fix)	Batch 12 (current)	Change
Total entities	5,773	2,583	Garbage filtered, dedup improved
Superseded claims	1,504 (36%)	24 (<1%)	-98% (SINGLE_VALUED_CLAIMS guard)
Avg confidence	0.978 (91% at 1.0)	0.428	Realistic after recalibration + decay
Hub-spoke topics (deg=1)	~794	~0	Eliminated by leaf-topic prune
Tests passing	87	117	+30 total
Validation errors	many	0	Strict Pydantic + coerce fallback
Entity resolution	N/A	P=1.0 R=0.72 F1=0.84	Gold-standard evaluation

The 24 superseded claims represent genuine temporal conflicts — e.g., different emails asserting different reporting lines for the same person. Multi-valued claim types (sent_to, mentioned, discussed) correctly remain ACTIVE. These conflicts are visible in the UI's conflict panel and in every context pack's conflicts field.